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Tarea Sección 5.5

3) $\int x^2 \sqrt{x^3+1}\,dx$ $\quad u = x^3+1 \quad du = 3x^2\,dx \quad \frac{du}{3x^2} = dx$

$\int x^2 \sqrt{u}\, \frac{du}{3x^2} = \frac{1}{3}\int u^{1/2} du = \frac{1}{3}\left(\frac{u^{3/2}}{\frac{3}{2}}\right) = \frac{u^{3/2}}{\frac{9}{2}}$

7) $\int x\sqrt{1-x^2}\,dx = \int x\sqrt{m}\,\frac{dm}{-2x} = -\frac{1}{2}\int \sqrt{m}\,dm = -\frac{1}{2}\int m^{1/2} dm =$

$m = 1-x^2$

$dm = -2x\,dx$ $\quad = -\frac{1}{2}\left[\frac{m^{3/2}}{\frac{3}{2}}\right] = -\frac{m^{3/2}}{3} = -\frac{1}{3}(1-x^2)^{3/2} + C$

$\frac{dm}{-2x}$

9) $\int (3x-2)^{20} dx = \int (m)^{20}\frac{dm}{3} = \frac{1}{3}\int m^{20} = \frac{1}{3}\left(\frac{m^{21}}{21}\right) = \frac{m^{21}}{63}$

$m = 3x-2$

$dm = 3\,dx$ $\quad \frac{(3x-2)^{21}}{63} + C$

$\frac{dm}{3} = dx$

11) $\int \cos\left(\frac{\pi t}{2}\right) dt = \int \cos(m)\frac{dm}{\pi/2} = \frac{\pi}{2}\int \cos(m)\,dm = \frac{\pi}{2}\operatorname{sen}(m)$

$m = \frac{\pi t}{2}$ $\quad \frac{\pi}{2}\operatorname{sen}\left(\frac{\pi t}{2}\right) + C$

$dm = \frac{\pi}{2}dt$

$\frac{dm}{\frac{\pi}{2}} = dt$

13) $\int \operatorname{sen} \pi t\,dt = \int \operatorname{sen} m\,\frac{dm}{\pi} = \frac{1}{\pi}\int \operatorname{sen} m\,dm = \pi - \cos\pi$

$m = \pi t$

$dm = \pi\,dt$

$\frac{dm}{\pi} = dt$ $\quad = -\frac{1}{\pi}\cos \pi t + C$

15) $\int \cos^3\theta \operatorname{sen}\theta\,d\theta = \int m^3 \operatorname{sen}\theta\,\frac{dm}{-\operatorname{sen}\theta} = -\int m^3\,dm = -\frac{m^4}{4} =$

$m = \cos\theta$

$dm = -\operatorname{sen}\theta\,d\theta$ $\quad -\frac{\cos^4\theta}{4} + C$

$\frac{dm}{-\operatorname{sen}\theta} = d\theta$

17) $\int \frac{e^u}{(1-e^u)^2}\,du = \int \frac{e^u}{m^2}\frac{dm}{-e^u} = -\int m^{-2}\,dm = -\frac{m^{-1}}{-1}$

$m = 1-e^u$

$dm = -e^u\,du$

$\frac{dm}{-e^u} = du$

$= \frac{1}{m} = \frac{1}{1-e^u}$

19) $\int \frac{a+bx^2}{\sqrt{3ax+bx^3}}\,dx = \int \frac{a+bx^2}{\sqrt{m}}\frac{dm}{3a+3bx^2} = \int \frac{\cancel{a+bx^2}}{\sqrt{m}}\frac{dm}{3(\cancel{a+bx^2})} = \frac{1}{3}\int m^{-1/2}$

$m = 3ax+bx^3$

$dm = 3a+3bx^2\,dx$

$\frac{dm}{3a+3bx^2} = dx$

$= \frac{1}{3}\left(\frac{m^{1/2}}{\frac{1}{2}}\right) = \frac{1}{3}\,2m^{1/2} = \frac{2}{3}(ax+bx^3)+C$

21) $\int \frac{(\ln x)^2}{x}\,dx = \int \frac{m^2}{x}\frac{dm}{\frac{1}{x}} = \int m^2\,dm = \frac{m^3}{3} = \frac{1}{3}(\ln x)^3+C$

$m = \ln x$

$dm = \frac{1}{x}dx$

$\frac{dm}{\frac{1}{x}} = dx$

23) $\int \cos^4\theta \,\text{sen}\theta\,d\theta = \int m^4 \,\text{sen}\theta \frac{dm}{-\text{sen}\theta} = -\int m^4\,dm = \frac{m^5}{5}$

$m = \cos\theta$

$dm = -\text{sen}\theta\,d\theta$

$\frac{dm}{-\text{sen}\theta} = d\theta$

$= -\frac{1}{5}\cos^5\theta + C$

25) $\int e^x\sqrt{1+e^x}\,dx = \int e^x\sqrt{m}\frac{dm}{e^x} = \int m^{1/2} = \frac{m^{3/2}}{\frac{3}{2}} = \frac{2m^{3/2}}{3}$

$m = 1+e^x$

$dm = e^x\,dx$

$\frac{dm}{e^x} = dx$

$= \frac{2}{3}m^{3/2} = \frac{2}{3}(1+e^x)^{3/2}+C$

27) $\int (x^2+1)(x^3+3x)^4\,dx = \int \cancel{(x^2+1)}\,m^4\frac{dm}{3\cancel{(x^2+1)}} = \frac{1}{3}\int m^4\,dm = \frac{1}{3}\frac{m^5}{5}$

$m = x^3+3x$

$dm = 3x^2+3\,dx$

$\frac{dm}{3x^2+3} = dx$

$\frac{dm}{3(x^2+1)} = dx$

$= \frac{m^5}{15} = \frac{(x^3+3x)^5}{15}$

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29) $\int 5^t sen(5^t)\,dt = \int \cancel{5^t} sen(m) \frac{dm}{ln(5)\cancel{5^t}} = \frac{1}{ln(5)}\int sen(m)\,dm$

$m = 5^t$

$dm = ln(5)\,5^t\,dt$

$\frac{dm}{ln(5)5^t} = dt$

$= -\frac{1}{ln(5)} \cdot cos(m) = -\frac{1}{ln(5)} cos(5^t) + C$

31) $\int \frac{(arctan\,x)^2}{x^2+1}dx = \int \frac{m^2}{\cancel{x^2+1}} \cdot \cancel{x^2+1}\,dm = \int m^2\,dm = \frac{m^3}{3} = \frac{(tan^{-1}x)^3}{3} = \frac{1}{3}(tan^{-1}x)^3 + C$

$m = tan^{-1}x$

$dm = \frac{1}{x^2+1}dx \qquad \left.\frac{\frac{dm}{1}}{\frac{1}{x^2+1}}\right] \; x^2+1\,dm$

$\frac{dm}{\frac{1}{x^2+1}} = dx$

33) $\int cos(1+5t)\,dt = \int cos(m)\frac{dm}{5} = \frac{1}{5}\int cos(m)\,dm = \frac{1}{5}sen(m) = \frac{1}{5}sen(1+5t) + C$

$m = 1+5t$

$dm = 5\,dt$

$\frac{dm}{5} = dt$

35) $\int \sqrt{cot\,x}\,csc^2x\,dx = \int m^{1/2}\cancel{csc^2x}\frac{dm}{-\cancel{csc^2x}} = -\int m^{1/2}dm = -\frac{m^{3/2}}{\frac{3}{2}} = -\frac{2}{3}m^{3/2}$

$m = cot\,x$

$dm = -csc^2x\,dx \qquad \left.\frac{\frac{m}{1}}{\frac{3}{2}}\right]$

$\frac{dm}{-csc^2x} = dx$

$= -\frac{2}{3}(cot\,x)^{3/2} + C$

37) $\int senh^2(x) + cosh(x)\,dx = \int m^2 + \cancel{cosh\,x}\frac{dm}{\cancel{cosh\,x}} = \int m^2\,dm = \frac{m^3}{3}$

$m = senh\,x$

$dm = cosh\,x\,dx$

$\frac{dm}{cosh\,x} = dx$

$= \frac{1}{3}senh^3(x) + C$

39) $\int \frac{sen2x}{1+cos^2x}dx = \int \frac{sen2x}{m}\frac{dm}{sen2x} = -\int \frac{1}{m}dm = -ln|m| = -ln|1+cos^2x| + C$

$m = 1+cos^2x$

$dm = -2cosx\,senx\,dx = \int sen\,2x$

$\frac{dm}{-sen2x} = dx$

$$\frac{\frac{1}{1}}{\frac{\operatorname{sen} x}{\cos x}}$$

41) $\int \cot x \, dx = \int \frac{1}{\tan x} dx = \int \frac{1}{\frac{\operatorname{sen} x}{\cos x}} dx = \int \frac{\cos x}{\operatorname{sen} x} dx = \int \frac{\cancel{\cos x}}{U} \frac{dU}{\cancel{\cos x}}$

$U = \operatorname{sen} x$

$dU = \cos x \, dx$

$\frac{dU}{\cos x} = dx$

$= \int \frac{1}{U} dU = \ln|U| = \ln(|\operatorname{sen} x|) + C$

43) $\int \frac{dx}{\sqrt{1-x^2}\,\operatorname{sen}^{-1} x} = \int \frac{\sqrt{1-x^2}\, dm}{\sqrt{1-x^2}\, m} = \int \frac{1}{m} dm = \ln|m| = \ln(|\operatorname{sen}^{-1} x|) + C$

$m = \operatorname{sen}^{-1} x$

$dm = \frac{1}{\sqrt{1-x^2}} dx$

$\frac{dm}{\frac{1}{\sqrt{1-x^2}}} = dx$

45) $\int \frac{1+x}{1+x^2} dx = \int \frac{1 \, dx}{1+x^2} + \int \frac{x}{1+x^2} dx = \tan^{-1} x + \int \frac{\cancel{x}}{1+m} \frac{dm}{2\cancel{x}}$

$m = x^2$

$dm = 2x \, dx$

$\frac{dm}{2x} = dx$

$n = 1+m$

$dn = 1 dm$

$\frac{dn}{1} = dm$

$= \tan^{-1} x + \frac{1}{2} \int \frac{1}{1+m} dm = \tan^{-1} x + \frac{1}{2} \int \frac{1}{n} dn =$

$= \tan^{-1} x + \frac{1}{2} \ln|n| = \tan^{-1} x + \frac{1}{2} \ln(|1+x^2|) + C$

47) $\int x(2x+5)^8 dx = \int x \, m^8 \frac{dm}{2} = \int \frac{m-5}{2} \cdot m^8 \cdot \frac{dm}{2} = \frac{1}{4} \int m-5 \cdot m^8 \, dm$

$m = 2x+5$

$dm = 2 dx$

$\frac{dm}{2} = dx$

$= \frac{1}{4} \int m^9 - 5m^8 dm = \frac{1}{4} \frac{m^{10}}{10} - \frac{5m^9}{9}$

$m = 2x+5$

$\frac{m-5}{2} = x$

$= \frac{1}{4}\left(\frac{(2x+5)^{10}}{10} - \frac{5(2x+5)^9}{9}\right) = \frac{1}{40}(2x+5)^{10} - \frac{5}{36}(2x+5)^9 + C$

53) $\int_0^1 \cos\left(\frac{\pi t}{2}\right) dt = \int_0^1 \cos(m) \cdot \frac{2\,dm}{\pi} = \frac{2}{\pi}\int_0^1 \cos(m)\,dm$

$m = \frac{\pi t}{2}$

$dm = \frac{\pi}{2} t$

$2dm = \pi t$

$\frac{2dm}{\pi} = t$

$\frac{2}{\pi}\operatorname{Sen}(m)\Big|_0^1 = \frac{2}{\pi}\operatorname{Sen}\left(\frac{\pi t}{2}\right)\Big|_0^1 = \left[\frac{2}{\pi}\operatorname{Sen}\left(\frac{\pi \cdot 1}{2}\right)\right] - \left[\frac{2}{\pi}\operatorname{Sen}\left(\frac{\pi \cdot 0}{2}\right)\right] = \frac{2}{\pi}$

55) $\int_0^1 \sqrt[3]{1+7x}\,dx = \int_0^1 \sqrt[3]{m}\,\frac{dm}{7} = \frac{1}{7}\int_0^1 \sqrt[3]{m}\,dm = \frac{1}{7}\int_0^1 m^{1/3}\,dm = \frac{1}{7}\left(\frac{m^{4/3}}{\frac{4}{3}}\right)$

$m = 1+7x$

$dm = 7dx$

$\frac{dm}{7} = dx$

$= \frac{1}{7}\left(\frac{3m^{4/3}}{4}\right) = \frac{1}{7}\left(\frac{3}{4}(1+7x)^{4/3}\right)\Big|_0^1$

$= \left[\frac{1}{7}\left(\frac{3}{4}(1+7(1))^{4/3}\right)\right] - \left[\frac{1}{7}\left(\frac{3}{4}(1+7(0))^{4/3}\right)\right] = \frac{45}{28}$

57) $\int_0^{\pi/6} \frac{\operatorname{sen} t}{\cos^2 t}\,dt = \int_0^{\pi/6} \frac{\operatorname{sen} t}{m^2}\,\frac{dm}{-\operatorname{sen} t} = -\int_0^{\pi/6} \frac{1}{m^2}\,dm = -\int_0^{\pi/6} m^{-2}\,dm = -\frac{m^{-1}}{-1} = m^{-1}$

$m = \cos t$

$dm = -\operatorname{sen} t\,dt$

$\frac{dm}{-\operatorname{sen} t} = dt$

$= (\cos t)^{-1}\Big|_0^{\pi/6} = \left[\left(\cos\frac{\pi}{6}\right)^{-1} - (\cos 0)^{-1}\right] = 0.1547$

59) $\int_1^2 \frac{e^{1/x}}{x^2}\,dx = \int_1^2 \frac{m}{x^2}\,\frac{x^2\,dm}{-\frac{e^{1/x}}{x^2}} = \int_1^2 \frac{m}{x^2}\,\frac{x^2\,dm}{-e^{1/x}} = -\int_1^2 \frac{m}{e^{1/x}} \qquad -\int_1^2 \frac{m}{m}$

$m = e^{1/x}$

$dm - e^{1/x} \cdot \frac{d}{dx}\left(\frac{1}{x}\right) \qquad -\int_1^2 e^{1/x} \qquad b - a \qquad -\int_1^2 1$

$dm = -\frac{e^{1/x}}{x^2}\,dx \qquad -e^{1/2} - e^{1/1} \qquad -e - (-e) \qquad -m$

$\frac{dm}{-\frac{e^{1/x}}{x^2}} = dx \qquad \frac{\frac{dm}{1}}{-\frac{e^{1/x}}{x^2}} = -\frac{x^2\,dm}{e^{1/x}} \qquad e^{1/x}\Big|_1^2$

$e^{1/2} - e^{1/1} = e - \sqrt{e}$

61) $\int_{-\frac{\pi}{4}}^{\pi/4} (x^3 + x^4 \tan x)\, dx = \underline{0}$

63) $\int_0^{13} \frac{dx}{\sqrt[3]{(1+2x)^2}} = \int_0^{13} \frac{\frac{dm}{2}}{\sqrt[3]{m^2}} = \int_0^{13} \frac{1\, dm}{2\sqrt[3]{m^2}} = \frac{1}{2}\int_0^{13} m^{-2/3} = \frac{1}{2}\left(\frac{m^{1/3}}{\frac{1}{3}}\right)$

$m = 1+2x$

$dm = 2\, dx$

$\frac{dm}{2} = dx$

$= \frac{m^{1/3}}{\frac{2}{3}} = \frac{3}{2} m^{1/3} = \frac{3}{2}(1+2x)^{1/3} \Big|_0^{13} = \left[\frac{3}{2}(1+2(13))^{\frac{1}{3}}\right] - \left[\frac{3}{2}(1+2(0))^{\frac{1}{3}}\right] =$

$\frac{9}{2} - \frac{3}{2} = \underline{3}$

65) $\int_0^a x\sqrt{x^2+a^2}\, dx = \int_0^a x\sqrt{m}\, dx = \int_0^a x\, m^{1/2} \frac{dm}{2x} = \frac{1}{2}\int_0^a m^{1/2}\, dm = \frac{1}{2}\left(\frac{m^{3/2}}{\frac{3}{2}}\right)$

$m = x^2 + a^2$

$dm = 2x\, dx$

$\frac{dm}{2x} = dx$

$= \frac{1}{3} m^{3/2} = \frac{1}{3}(x^2+a^2)^{3/2}\Big|_0^a = \frac{1}{3}\left[(a^2+a^2)^{3/2} - (0^2+a^2)^{3/2}\right]$

$= \frac{1}{3}\left[(2a^2)^{3/2} - (a^2)^{3/2}\right] = \underline{\frac{1}{3}\left[2a\sqrt{2a} - a^3\right]}$

67) $\int_1^2 x\sqrt{x-1}\, dx = \int_1^2 x\sqrt{m}\, dm = \int_1^2 (1+m)\sqrt{m}\, dm = \int_1 (1+m)\, m^{1/2}\, dm$

$m = x-1 \qquad 1+m = x$

$dm = dx$

$\int_1^2 m^{1/2} + m^{3/2} = \frac{m^{3/2}}{\frac{3}{2}} + \frac{m^{5/2}}{\frac{5}{2}}\, dm$

$= \frac{2}{3} m^{3/2} + \frac{2}{5} m^{5/2} = \frac{2}{3}(x-1)^{3/2} + \frac{2}{5}(x-1)^{5/2}\Big|_1^2$

$= \left[\frac{2}{3}(2-1)^{3/2} + \frac{2}{5}(2-1)^{5/2}\right] - \left[\frac{2}{3}(1-1)^{3/2} + \frac{2}{5}(1-1)^{5/2}\right] =$

$\frac{16}{15} - 0 = \underline{\frac{16}{15}}$

$$69)\ \int_e^{e^4} \frac{dx}{x\sqrt{\ln x}} = \int_e^{e^4} \frac{1}{x\sqrt{m}}\, x\, dm = \int_e^{e^4} \frac{1}{\sqrt{m}}\, dm = \int_e^{e^4} m^{-1/2}\, dm = \frac{m^{1/2}}{1/2} = 2m^{1/2}$$

$m = \ln x$

$dm = \frac{1}{x}\, dx$

$\frac{dm}{\frac{1}{x}} = dx$

$$2(\ln x)^{1/2}\Big|_e^{e^4} = [2(\ln e^4)^{1/2}] - [2(\ln e)^{1/2}] = 4 - 2 = 2$$

$$71)\ \int_0^1 \frac{e^z+1}{e^z+z}\, dx = \int_0^1 \frac{e^z+1}{m}\, \frac{dm}{e^z} = \int_0^1 \frac{1}{m}\, dm = \ln|m| = \ln|e^z+z|\Big|_0^1$$

$m = e^z + z$

$dm = e^z\, dz$

$\frac{dm}{e^z} = dz$

$$= \ln|e^1+1| - \ln|e^0+0| = \ln|e^1+1|$$

$$73)\ \int_0^1 \frac{dx}{(1+\sqrt{x})^4} = \int_0^1 \frac{1}{m^4}\, \frac{2\, dm}{x^{-1/2}} = \int_0^1 \frac{1}{m^4}\, 2\, x^{1/2}\, dm$$

$m = 1 + \sqrt{x} = 1 + x^{1/2}$

$dm = \frac{1}{2} x^{-1/2}\, dx$

$\frac{dm}{\frac{1}{2} x^{-1/2}} = dx$

$$= 2\int_0^1 \frac{1}{m^4} \cdot m-1\, dm = 2\int_0^1 \frac{m-1}{m^4}\, dm$$

$$= 2\left(\int_0^1 \frac{m}{m^4}\, dm - \int_0^1 \frac{1}{m^4}\, dm\right)$$

$1 + x^{1/2} = m$

$x^{1/2} = m - 1$

$$= 2\left(\int_0^1 \frac{1}{m^3}\, dm - \int_0^1 \frac{1}{m^4}\, dm\right)$$

$m(1) = 1 + \sqrt{1} = 2$

$m(0) = 1 + \sqrt{0} = 1$

$$= 2\left(\int_0^1 m^{-3}\, dm - \int_0^1 m^{-4}\, dm\right)$$

$$= 2\left(\frac{m^{-2}}{-2} - \frac{m^{-3}}{-3}\right)\Big|_1^2$$

$$= \left[2\left(-\frac{2^{-2}}{2} + \frac{2^{-3}}{3}\right)\right] - \left[2\left(-\frac{1^{-2}}{2} + \frac{1^{-3}}{3}\right)\right]$$

$$= -\frac{1}{6} - \left(-\frac{1}{3}\right) = \frac{1}{6}$$

77) $\int_{-2}^{2}(x+3)\sqrt{4-x^2}\,dx = \int_{-2}^{2} x\sqrt{4-x^2} + 3\sqrt{4-x^2}\,dx$

$\int_{-2}^{2} x\sqrt{4-x^2}\,dx = \int_{-2}^{2} x\sqrt{m}\,\frac{dm}{-2x} = -\frac{1}{2}\int_{-2}^{2}\sqrt{m}\,dm = -\frac{1}{2}\int_{-2}^{2} m^{1/2}\,dm$

$m = 4-x^2$

$dm = -2x\,dx$

$\frac{dm}{-2x} = dx$

$= -\frac{1}{2}\left(\frac{m^{3/2}}{\frac{3}{2}}\right) = -\frac{m^{3/2}}{3} = -\frac{1}{3}m^{3/2}\Big|_0^0$

$= \left(-\frac{1}{3}\cdot 0^{3/2}\right) - \left(-\frac{1}{3}\cdot 0^{3/2}\right) = 0$

$m(2) = 0$

$m(-2) = 0$

$3\sqrt{4-x^2} = 0$

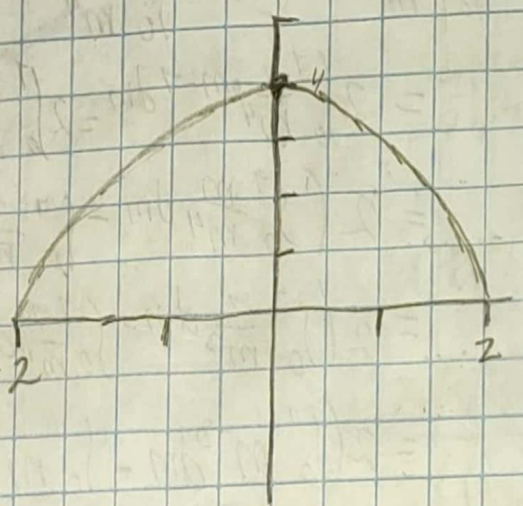
$4-x^2 = 0$

$-x^2 = -4$

$x = \pm\sqrt{4}$

$x = -2$

$x = 2$



$3\int_{-2}^{2}\sqrt{4-x^2}\,dx =$

$\frac{\pi r^2}{2} = \frac{2^2\pi}{2} = \frac{4\pi}{2} = 2\pi \quad 3\cdot 2\pi = 6\pi$

81) $t = 0$ $\quad r(t) = 100e^{-0.01t}$ litros/minuto

$\int_0^{60} 100e^{-0.01t}\,dt = 100\int_0^{60} e^m \frac{dm}{-0.01} = \frac{100}{-0.01}\int_0^{60} e^m\,dm$

$m = -0.01t$

$dm = -0.01\,dt$

$\frac{dm}{-0.01} = dt$

$= \frac{100}{-0.01}(e^m) = \frac{100}{-0.01}e^{-0.01t}\Big|_0^{60}$

$\left[-\frac{100}{0.01}e^{-0.01(60)}\right] - \left[-\frac{100}{0.01}e^{-0.01(0)}\right] = 4{,}511{,}88 \approx 4{,}512$